Worked Example 7.1

Calculate the terminal velocity of a raindrop of radius 0.2 cm. (Density of water 1000 kg/m^3 and that of air 1 kg /m^3).

Solution:

Step 1:

r = 0.2 cm

$V_t = ?$

Step 2:

$$v_t = \frac{2gr^2}{9\eta}\,(\rho - \sigma)$$

Step 3:

$$v_t = \frac{2\times 9.8\times (0.2^{-3})^2 \times 999}{9\times 10^{-3}}$$

$V_t = 8.7$ m/s

7.3 Fluids in Motion:

Fluids can move or flow in many ways. Water may flow smoothly and slowly in a quiet stream or violently over a water fall. The air may form a gentle breeze or a raging tornado. To deal with such diversity, it is necessary to classify some of the basic types of fluid flows.

7.3.1 Types of Fluids:

In ***steady flow*** the velocity of the fluid particles at any instant is constant as time passes. Every particle passing through certain point has the same velocity whereas at another location the velocity may vary, or as in river, which usually flows fastest near its center and slowed near its banks.

Streamline flow:

When a fluid flows slowly as shown along a pipe, the flow is said to be steady and lines are called streamlines. In figure (a) 7.2, they are parallel to the walls of the pipe. A streamline is a line drawn in the fluid such that a tangent to the stream – line at any point is parallel to the fluid velocity at that point. The fluid velocity can vary (in both magnitude and direction) from point to point along a streamline, but at any given point, the velocity is constant in time, as required by the condition of steady flow. Such steady flow of fluid is called ***streamline flow OR Laminar flow***

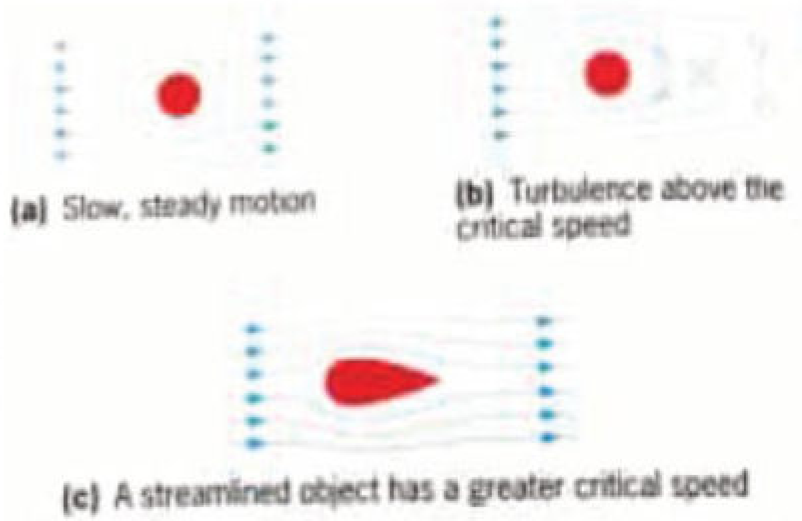


Fig: 7.2 types of fluid flow

Laminar flow refers to the smooth, orderly movement of a fluid in which layers of the fluid slide past each other in parallel. In laminar flow, the fluid moves in well-defined streamlines without any significant mixing or turbulence. This type of flow is characterized by a low Reynolds number, indicating a relatively slow and viscous flow.

Key characteristics of laminar flow include:
- Streamline Flow
- Smooth Velocity Profile
- Low Mixing and Diffusion
- Low Shear Stress
- Predictable Flow Behavior

Laminar flow is commonly observed in situations with low flow rates, small pipe diameters, and high fluid viscosity. It can be found in applications such as certain laboratory experiments, microfluidic devices, and some industrial processes that require precise control of fluid motion.

Unsteady flow:

Unsteady flow refers to the flow of a fluid that changes with time. In unsteady flow, the properties of the fluid, such as velocity, pressure, and density, vary at different locations and change over time. This is in contrast to steady flow, where the fluid properties remain constant at any given location.

Unsteady flow can occur in various fluid systems and is often associated with dynamic or transient conditions. Some examples of situations that involve unsteady flow include:

Water waves, Pulsatile blood flow, Pipe filling and draining, Turbulent flow etc.

Understanding unsteady flow is crucial in various engineering applications, such as the design of pipelines, pumps, turbines, and environmental fluid dynamics. It allows engineers and scientists to study the transient behavior of fluids and analyze the effects of time-varying flow conditions on system performance and stability

Incompressible fluid flow:

Mostly liquids are incompressible during which density of fluid remains constant even with varying pressure. In contrast, gases are highly compressible. However, there are certain situations in which the density of a flowing gas remains constant enough that the flow can be considered incompressible.

Non-viscous fluid flow:

An incompressible, non-viscous fluid is called an ***ideal fluid***, with zero viscosity flows in an unhindered manner with no dissipation of energy. Although no real fluid has zero viscosity at normal temperatures, some fluids have negligibly small viscosities.

7.3.2 Transition from Laminar to turbulent flow:

In laminar flow, the streamlines of a fluid follow smooth paths. In contrast, for a fluid in turbulent flow, vortices form, detach and, propagate. When viscous or laminar flow exceeds certain limit is said to be transforming from laminar to turbulent. Just look at cigarette smoke undergoes a transition from laminar to turbulent flow. What is the criterion that determines whether flow is laminar or turbulent?

In 19th century Reynolds investigated the conditions that would give turbulence in the flow of a fluid. ***Reynolds number***, R_e, which is ***the ratio of the typical inertial force to the viscous force and thus is a pure dimensionless number.*** The inertial force has to be proportional to the density, ρ, and the typical velocity of the fluid, $\overline{v}$, because of rate of change of momentum i.e. $F = \frac{dp}{dt}$. The viscous force is proportional to the viscosity 'η' and inversely proportional to the characteristic length scale 'L' over which the flow varies. For flow through a pipe with a circular cross-section, this length scale is the diameter of the pipe, L= 2r. Thus, the formula for calculating the Reynolds number is

DO YOU KNOW?

Make a small hole near the bottom of an open tin can. Fill the can with water, which then proceeds to spurt from the hole. If you cover the top of the can firmly with the palm of your hand, the flow stops.

$$\text{Kinematics Viscosity} = \frac{\text{Dynamic Viscosity}}{\text{Fluid density}}$$

$$\text{Rynollds no} = \frac{\text{Fluid velocity} \times \text{Internal diameter}}{\text{Kinematic Viscosity}}$$

$$R_e = \frac{\rho \overline{v} L}{\eta} \;\ldots\ldots(7.4)$$

The velocity of liquid flow is given by

$$V = \frac{R_e \eta}{2\rho} \;\ldots\ldots(7.5)$$

V= Speed of fluid
r = radius
η= Viscosity
ρ=fluid density
R_e = Reynolds Number

The Reynolds number is important in analyzing the type of flow, when there is substantial velocity gradient. The flow is determined by using following standards:

. **Laminar** when *Re < 2300*

. **Transient** when *2300 < Re < 4000*

Transient flow, is flow where the flow velocity and pressure are changing with time. When changes occur to a fluid system such as the starting or stopping of a pump, closing or opening a valve, or changes in tank levels, then transient flow conditions exist: otherwise the system is steady state.

. **Turbulent** when *4000 < Re*

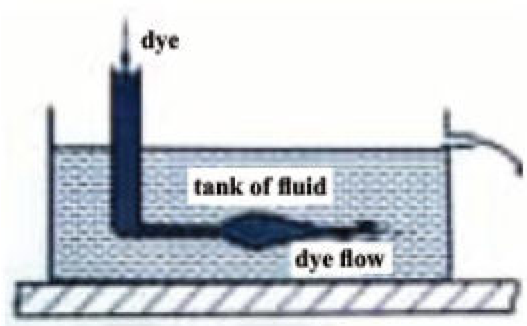
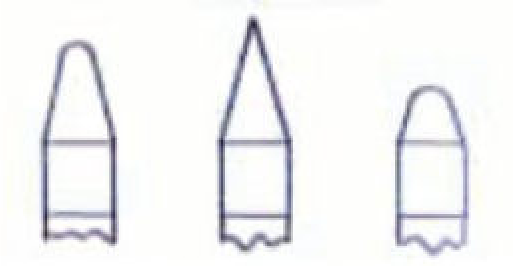


Fig: 7.3

The flow of liquid can easily be demonstrated with the apparatus shown in figure 7.3. A dye flows from the tube into the tank and by altering the pressure head the flow can be made either turbulent or laminar. As you can see in figure 7.3, the flow becomes turbulent when the line ceases to be straight.

The streamlining of bodies are most important in the design of cars, submarines and nose cones of aircrafts and rockets, since a reduction in drag can reduce vibration and also save large amounts of fuel. Figure 7.4 shows the best shapes for rocket cones for the ***subsonic***, ***supersonic*** and ***hypersonic*** flight respectively.



subsonic *supersonic* *hypersonic*

Fig: 7.4

Worked Example 7.2

The volume rate of an air conditioning system to be 3.84×10^{-3} m^3/s. The air is sent through an insulated, round conduit with a diameter of 18 cm. This calculation assumed laminar flow. (a) Was this a good assumption? (b) At what velocity would the flow become turbulent?

Solution:

Step 1:

Volume rate = 3.84×10^{-3} m^3/s.

η of air = 0.0181 m Pa.s

ρ of air = 1.23 kg/m^3

diameter = 18 cm

Velocity =?

Step 2:

Volume rate = Av

$3.84\times10^{-3} = \pi\,(0.09)\,v$

$v = 0.15$ m/s

Step 3:

$$R_e = \frac{2\rho v r}{\eta}$$

$$R_e = \frac{2\times(1.23)\times0.15\times0.09}{0.0181\times10^{-3}}$$

$R_e = 1835$

Since the Reynolds number is 1835 < 2000, the flow is laminar and not turbulent. The assumption that the flow was laminar is valid.

Step 4:

To find the maximum speed of the air to keep the flow laminar, consider the Reynolds number

$$R_e = \frac{2\rho\ \ r}{\eta} \leq 2000$$

$$v = \frac{2000(0.0181\times10^{-3})}{2(1.23)(0.09)}$$

$$v = 0.16\,m/s$$

Significance:

When transferring a fluid from one point to another, it desirable to limit turbulence. Turbulence results in wasted energy, as some of the energy intended to move the fluid is dissipated when eddies are formed. In this case, the air conditioning system will become less efficient once the velocity exceeds 0.16 m/s, since this is the point at which turbulence will begin to occur.

7.3.3 Fluid flow and resistance to motion in fluids involve turbulent rather than laminar conditions:

In practical examples of fluid flow and resistance to motion in fluids, turbulent conditions are more prevalent than laminar conditions. Turbulent flow is a type of fluid motion characterized by chaotic and irregular movement of fluid particles, resulting in whirlpool, fluctuations in velocity and pressure. On the other hand, laminar flow is smooth, ordered, and occurs in layers without any significant mixing or swirling.

Several factors contribute to the prevalence of turbulent flow in practical scenarios:

High Reynolds Numbers: Turbulent flow is more likely to occur at higher Reynolds numbers, which is a dimensionless parameter that measures the ratio of inertial forces to viscous forces in the fluid. In many real-world applications, such as in industrial processes, transportation, and natural phenomena like rivers and atmospheric flows, the Reynolds numbers are often large enough to induce turbulent behavior.

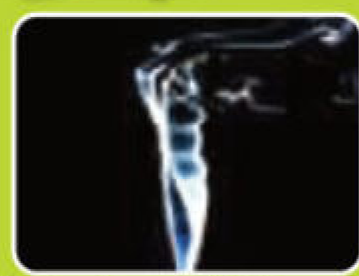
Rough Surfaces: When fluid flows over rough surfaces, such as in pipes, channels, or around objects, it can lead to turbulent flow due to the disruption of the fluid layers. Turbulence enhances the mixing of fluid components and affects the resistance to motion.

High Velocities: High flow velocities can promote turbulent behavior, especially in situations where the fluid encounters sudden changes in cross-sectional area or flow direction as shown in figure b.

Agitation and Stirring: In industrial processes, mixing, and agitation systems, turbulent flow is deliberately induced to ensure efficient mixing of substances and heat transfer.

Pressure Gradients: Rapid changes in pressure along the flow path can lead to turbulent flow patterns, as the fluid tries to adjust to the varying conditions.

DO YOU KNOW?



Smoke rises smoothly for a while and then begins to form swirls and eddies. The smooth flow is called laminar flow, whereas the swirls and eddies typify turbulent flow. Smoke rises more rapidly when flowing smoothly than after it becomes turbulent, suggesting that turbulence poses more resistance to flow.

Fig: 7.5

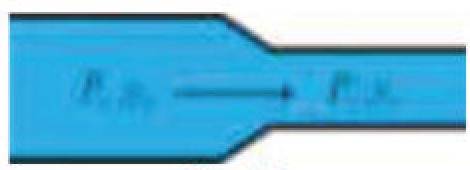


Fig: 7.6

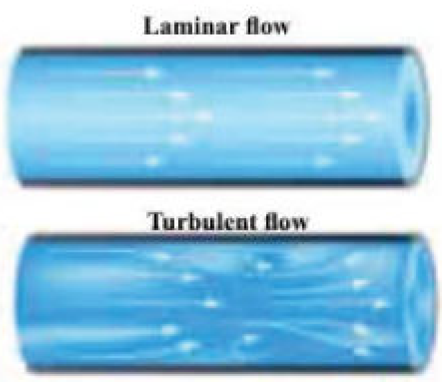


Fig: 7.7

Examples of practical situations involving turbulent flow include:

- Airflow around vehicles, airplanes, and wind turbines as depicted in figure a.
- Fluid flow in pipes, especially in scenarios with high flow rates or rough internal surfaces as shown in figure c.
- Rivers and streams, where the irregularities in the bed and banks induce turbulence.
- Ocean currents and waves, which often involve turbulent motion.
- Mixing processes in chemical reactors, industrial tanks, and bioreactors.
- Combustion in engines and furnaces, where turbulent mixing of fuel and air improves combustion efficiency.

While laminar flow can occur in specific situations, such as slow and smooth flow in small tubes or in some laboratory setups, turbulent flow dominates in most real world fluid flow applications due to its complex and dynamic behavior, which significantly influences resistance to motion and various other fluid phenomena.

7.4 Equation of Continuity:

Suppose a steady laminar flow of a fluid through an enclosed tube or pipe as shown in figure the speed of the fluid varies with the diameter of the tube variation. The mass flow rate which follows the ***law conservation of mass*** is defined as **the mass Δm of fluid that passes at given point per unit time Δt**:

mass flow rate $= \frac{\Delta m}{\Delta t}$ ……(7.6)

In figure 7.8, the volume of a fluid passing through area A_1 in a time Δt is $A_1 \Delta l_1$, where Δl_1 is the distance the fluid moves in time Δt. The velocity of fluid passing through A_1 is $v_1 = \Delta m_1 / \Delta t$. Then the mass flow rate:

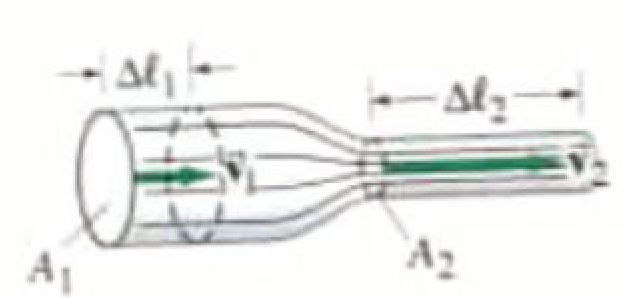


Fig: 7.8

$$\frac{\Delta m_1}{\Delta t} = \frac{\rho_1 \Delta V_1}{\Delta t}$$
$$= \frac{\rho_1 A_1 \Delta l_1}{\Delta t} = \rho_1 A_1 v_1$$

Where $\Delta V_1 = A_1 \Delta l_1$ is the volume of mass Δm. Similarly, through A_2, the flow rate is $\rho_2 A_2 v_2$. Since no fluid flows in or out the sides of the tube, the flow rates through A_1 and A_2 must be equal.

$$\frac{\Delta m_1}{\Delta t} = \frac{\Delta m_2}{\Delta t} \quad ……(7.7)$$

and

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

This is called the equation of continuity.

If the fluid is incompressible which is an excellent approximation for liquids under most circumstances, then $\rho_1 = \rho_2$, the equation of continuity is

$A_1 V_1 = A_2 V_2$ (ρ is constant) ……(7.8)

The product AV is the volume rate of flow which is defined as ***the volume of fluid passing a given point per second.***

$$\frac{\Delta v}{\Delta t} = A\frac{\Delta l}{\Delta t} = Av \quad \ldots\ldots(7.9)$$

Its SI unit is m^3/s.

The product of area and velocity is describing that where the cross-sectional area is large, the velocity is small, and where the area is small, the velocity is large.

Self-Assessment Questions:

1. State the equation of continuity in terms of fluid flow.
2. How does the equation of continuity relate to conservation of mass?
3. How does the velocity of a fluid change when it flows through a constricted pipe?

Worked Example 7.3

The radius of aorta is about 1.2 cm, and the blood passing through it has a speed of about 40 cm/s. A typical capillary has a radius of about 4×10^{-4} cm, and blood flows through it at a speed of about 5×10^{-4} m/s. Estimate the number of capillaries that are in the body.

Step 1:

radius of aorta = 1.2 cm
speed in = 40 cm/s
Capillary radius = 4×10^{-4} cm
Speed out = 5×10^{-4} m/s
No of capillaries =?

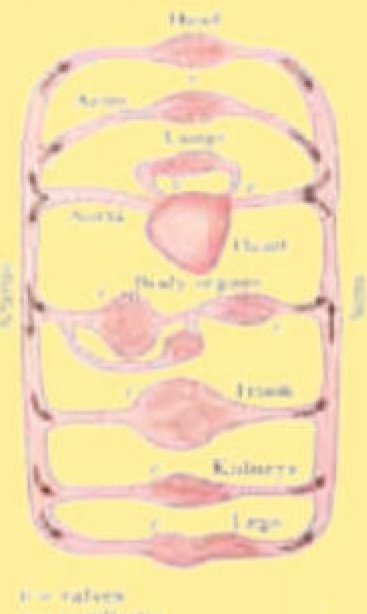
Let A_1 be the area of the aorta and A_2 be the area of all the capillaries through which blood flows. Then $A_2 = N\pi r_{cap}^2$, where $r_{cap} = 4\times10^{-4}$ cm is the estimated average radius of one capillary. From equation of continuity, we have

$$v_2 A_2 = v_1 A_1$$
$$v_2\, N\, r_{cap}^2 = v_1\ r_{aorta}^2$$
$$N = \frac{v_1}{v_2}\,\frac{r_{aorta}^2}{r_{cap}^2}$$

$N = (0.40\ m/s\ /\ 5\times10^{-4})\ (1.2\times10^{-2}\ m\ /\ 4\times10^{-6}\ m)^2$

$N = 7\times10^9$

on the order of billion capillaries.



Assume that blood density is same from aorta to capillaries. By equation of continuity, the volume flow rate in the aorta must equal the volume flow rate through all the capillaries. The total area of all the capillaries is given by the area of a typical capillary multiplied by the total number N of capillaries.

7.5 Bernoulli's Principle:

When a fluid is in motion the pressure within the fluid flow varies with velocity of the fluid for streamlined flow. This pressure variation is a consequence of Bernoulli's theorem proposed in 1740. Bernoulli's principle states that *the velocity of a fluid is high, the pressure is low, and the velocity is low, the pressure is high.*

DO YOU KNOW?

When a truck moves very fast, it created a low pressure area, so dusts are being pulled along in the low pressure area. If we stand very close to railway track in the platform, when a fast train passes us, we get pulled towards the track because of the low pressure area generated by the sheer speed of the train.

7.5.1 Bernoulli's Equation:

Bernoulli's equation deals with energy conservation law for the steady-state flow of incompressible fluids, such as water. It relates the energy of the fluid in terms of its pressure, velocity, and height.

Bernoulli's equation can be derived from the first principle using the law of conservation of energy. According to energy conservation principles, energy can neither be created nor destroyed. Therefore, during streamline flow, the total mechanical energy remains constant. A few assumptions need to be made before deriving the equation.

Assumptions

- The flow must be steady and streamline.
- The fluid is incompressible the density should remain constant at all points during the flow.
- There are no viscous forces in the fluid, and friction is negligible.

Consider a pipe whose diameter and elevation change as a fluid passes through it. Consider a small mass of the fluid with density ρ that flows from point 1 to 2 as shown in figure 7.9. The work done by a force F on the fluid to displace it by an infinitesimal distance Δx is given by,

$$W = F\Delta x$$

Therefore, at points 1 and 2, the work done are,

$$\Delta W_1 = F_1 \Delta x_1$$
$$\Delta W_2 = F_2 \Delta x_2$$

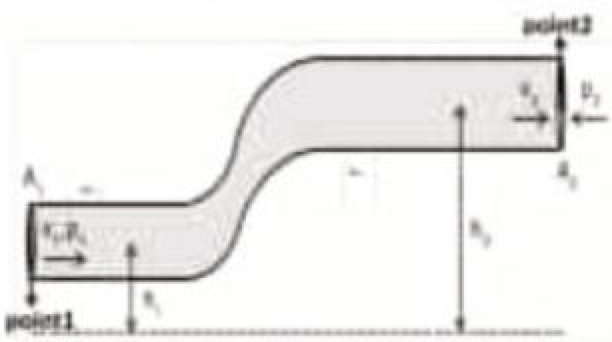


Fig: 7.9 fluid passthrough pipe

Total work done when the fluid moves 1 to 2 is,

$$\Delta W = \Delta W_1 - \Delta W_2$$
$$\text{or,} \; \Delta W = F_1 \Delta x_1 - F_2 \Delta x_2$$

Force is the product of pressure and area $(F = pA)$, and the volume is the product of length and cross − sectional area $(V = Ax)$. Therefore,

$$\Delta W = p_1A_1\Delta x_1 - p_2A_2\Delta x_2 = p_1\Delta V - p_2\Delta V = (p_1 - p_2)\Delta V$$

Now, when the fluid moves from point 1 to 2, there is a change of kinetic energy.

$$\Delta K.E = \frac{1}{2} m_2v_2^2 - \frac{1}{2} m_1v_1^2 = \frac{1}{2}\rho\Delta V v_2^2 - \frac{1}{2}\rho\Delta V v_1^2 = \frac{1}{2}\rho\Delta V(v_2^2 - v_1^2) \text{ ......(7.10)}$$

Similarly, the change in potential energy is given by,

$$\Delta U = mgh_2 - mgh_1 = \rho\Delta Vg(h_2 - h_1) \text{ ......(7.11)}$$

The work done in moving the fluid is the sum of the change in kinetic and potential energies.

$$\Delta W = \Delta K.E + \Delta U$$

$$\text{or, } (p_1 - p_2)\Delta V = \frac{1}{2}\rho\Delta V(v_2^2 - v_1^2) + \rho\Delta Vg(h_2 - h_1)$$

$$\text{or, } p_1 - p_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(h_2 - h_1)$$

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

$$\text{or, } p + \frac{1}{2}\rho V^2 + \rho gh = \text{constant}$$

This is Bernoulli's equation.

Therefore

$P + ½\rho V^2 + \rho g h = \text{Constant}$ ......(7.12)

The height h is measured from some convenient reference level, for example at the surface of the liquid. The equation is really a statement of conservation of energy per unit volume in the fluid.

- $½\rho V^2 + \rho g h$ is total pressure.
- $\rho g h$ is static pressure.
- $½\rho V^2$ is dynamic pressure.

Worked Example 7.4

Water leaves the jet of a horizontal hose at 10 m/s, if the velocity of water within the hose is 0.4 m/s, calculate the pressure P within the hose. (Density of water 1000 kg/m^3 and atmospheric pressure is 100000 Pa).

Solution:

Step 1:

V_1 = 10 m/s

V_2 = 0.4 m/s

P = ?

Density of water = 1000 kg/m^3

Atmospheric Pressure = 100000 Pa

Step 2:

Here $h_1 = h_2$, so Bernoulli's equation becomes

$P_1 + ½\rho V_1^2 = P_2 + ½\rho V_2^2$

Step 3:

$100000 + ½ \times 1000 \times 100 = P_2 + ½ \times 1000 \times 0.16$

$P_2 = 1.5 \times 10^5$ Pa

7.5.2 Applications of Bernoulli's Principle:

Bernoulli's principle, also known as the Bernoulli effect, describes the relationship between fluid velocity and pressure in a flowing fluid. It states that in an ideal, incompressible, and non-viscous fluid, the sum of the fluid's kinetic energy (velocity) and potential energy (pressure) remains constant along a streamline. This principle has various practical applications, including those mentioned in your question:

Filter Pump: In a filter pump, Bernoulli's principle is applied to increase the pressure of the fluid passing through the pump. By reducing the cross-sectional area of the pump's outlet, the fluid's velocity increases according to the principle, leading to a decrease in pressure (as kinetic energy increases). This pressure drop helps draw fluid into the pump and through the filter medium, facilitating the filtration process.

Fig: 7.10 Filter pump

Venturi Meter: A Venturi meter is a device used to measure the flow rate of a fluid in a pipe. It consists of a gradually narrowing tube (Venturi tube) inserted in the pipe. As the fluid flows through the narrowing section, its velocity increases according to Bernoulli's principle, and the pressure decreases. By measuring the pressure difference between the narrowest section and the wider parts of the pipe, the flow rate can be determined.

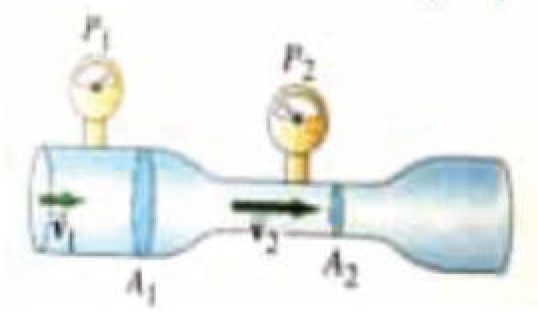


Fig: 7.11 Venturi meter

Atomizer: Atomizers are devices used to convert liquids into fine sprays or mists. The application of Bernoulli's principle is crucial in this process. As the liquid passes through a small nozzle in the atomizer, its velocity increases, leading to a decrease in pressure according to the principle. This decrease in pressure facilitates the breakup of the liquid into tiny droplets or a fine spray.

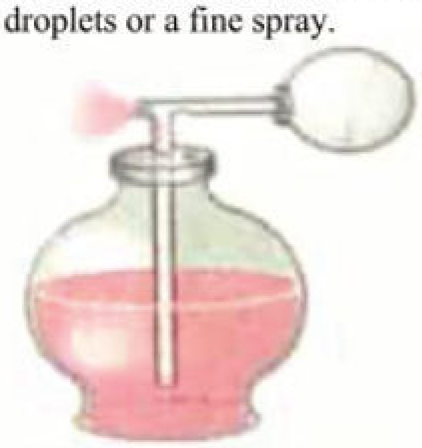


Fig: 7.12 Atomizer

DO YOU KNOW?

Why does smoke go up a chimney?

It's partly because hot air rises as its less dense and therefore buoyant. When wind blows across the top of a chimney, the pressure is less there than inside the house. Hence, air and smoke are pushed up the chimney by the higher indoor pressure. Even on an apparently still night there is usually enough ambient air flow at the top of a chimney to assist upward flow of smoke.

Flow of Air over an Aerofoil: When air flows over an aerofoil (such as the wing of an aircraft), the shape of the aerofoil causes the air to travel faster over the top surface compared to the bottom surface. According to Bernoulli's principle, the air pressure decreases over the top surface due to the increased velocity, creating a pressure difference that results in lift, allowing the aircraft to stay airborne

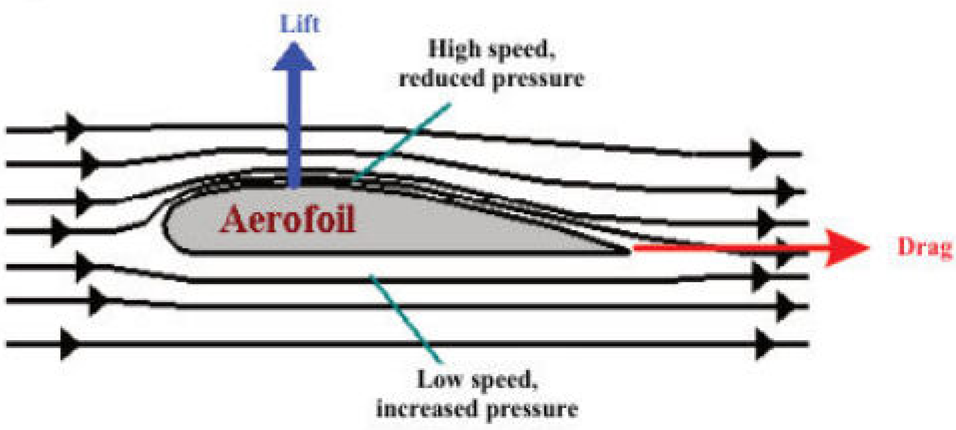


Fig: 7.13 An aerofoil

Blood Physics: In the circulatory system, Bernoulli's principle is applicable to the flow of blood through blood vessels, particularly in areas of constriction or stenosis. When the diameter of a blood vessel narrows, the blood velocity increases, leading to a decrease in pressure, which can have clinical implications in conditions such as atherosclerosis or stenosis.

It's essential to note that while Bernoulli's principle is an excellent theoretical tool for understanding fluid behavior in these applications, real-world fluids may have additional complexities, such as viscosity and compressibility, which need to be considered for precise calculations and analyses.

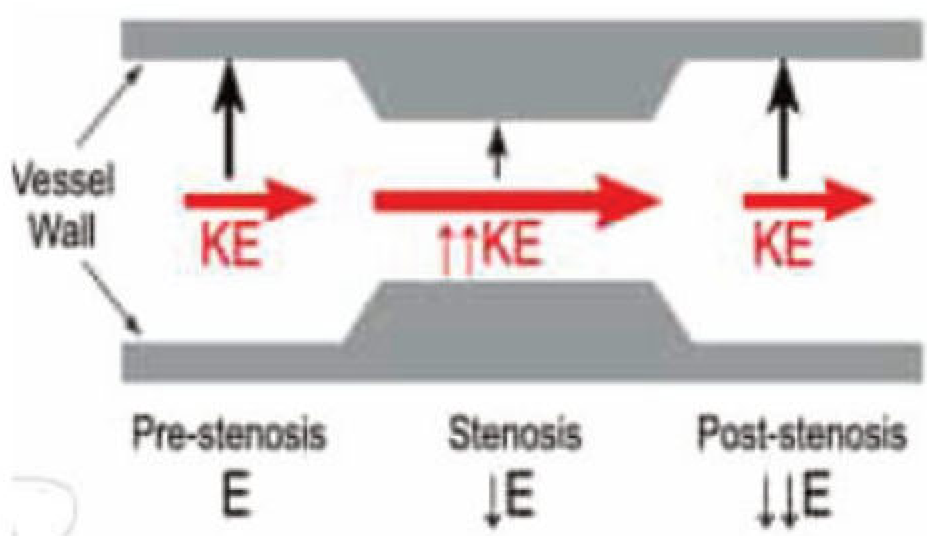
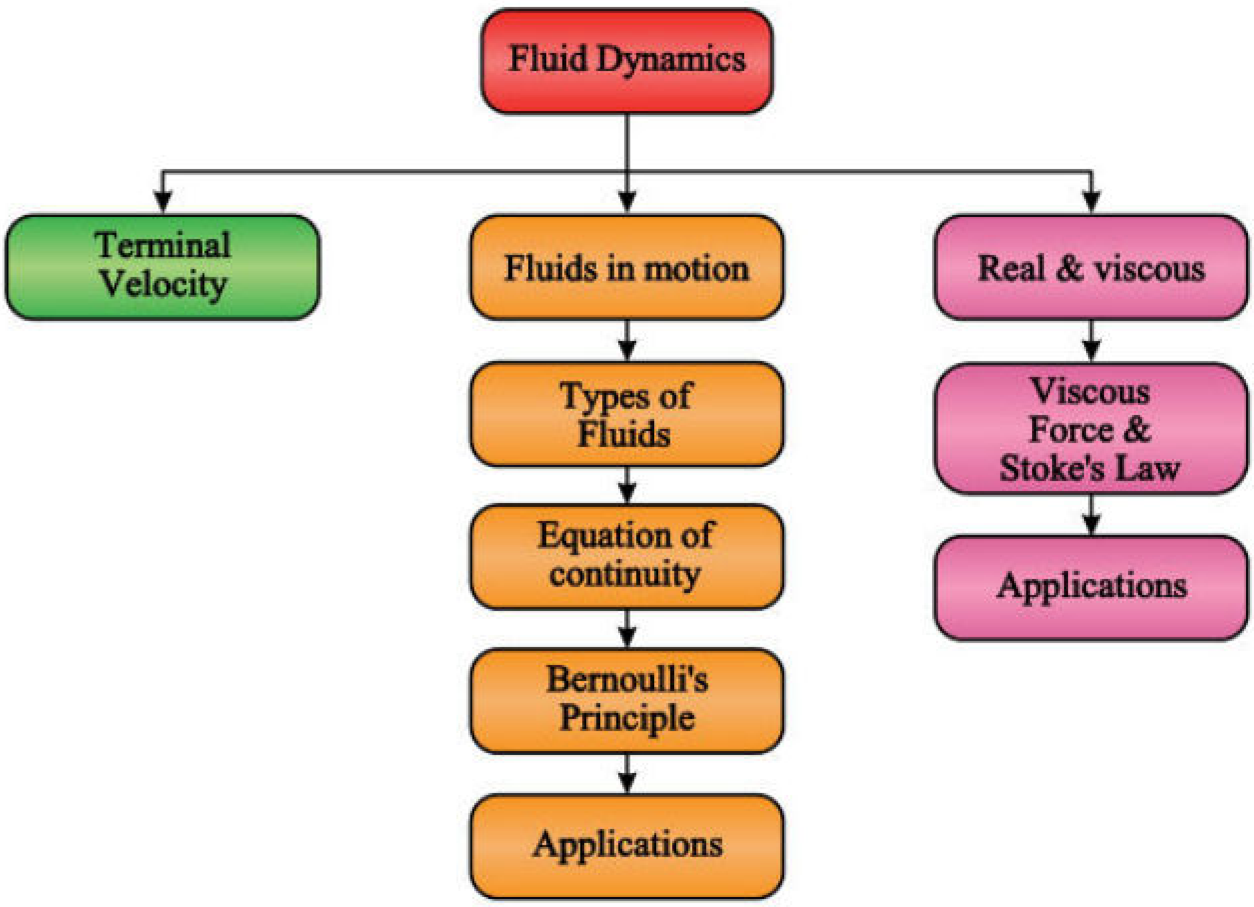


Fig: 7.14
Stenosis or pre stenosis of blood vessels walls

SUMMARY

- The physics of fluids in motion is Fluid Dynamics:
- A frictional force between adjacent layers of fluid as the layers move past one another is called Viscosity:
- The rate of change of velocity with distance normal to the direction of flow of the layers of the fluid with respect to object passing through the fluid is known as Velocity Gradient:
- When fluid resistance of a falling object equals its weight, the net force is zero and no further acceleration occurs is called Terminal Velocity:
- The velocity of the fluid particles at any is constant as time passes is called Steady flow: Whenever the velocity at a point in the fluid changes as time passes, such flow becomes turbulent is known as unsteady flow:
- An incompressible, non-viscous fluid is called an ***ideal fluid***,
- The sliding of layers of fluid parallel and smoothly to direction of flow is Laminar Flow:
- The ratio of the typical inertial force to the viscous force and thus is a pure dimensionless number is called Reynolds Number:
- The velocity of a fluid is high, the pressure is low, and the velocity is low, the pressure is high is known as Bernoulli's Principle:

CONCEPT MAP


Fluid Dynamics
Terminal Velocity
Fluids in motion
Types of Fluids
Equation of continuity
Bernoulli's Principle
Applications
Real & viscous
Viscous Force & Stoke's Law
Applications

EXERCISE

Section (A): Multiple Choice Questions (MCQs)

1. For an incompressible fluid, the flow rate is
   a) equal for all surfaces.
   b) constant throughout the pipe.
   c) greater for the larger parts of the pipe.
   d) none of the above

2. Bernoulli's principle states that for horizontal flow of a fluid through a tube, the sum of the pressure and energy of motion per unit volume is
   a) increasing with time
   b) decreasing with time
   c) constant
   d) varying with time

3. Which of the following is associated with the law of conservation of energy in fluids?
   a) Archimedes' principle
   b) Bernoulli's principle
   c) Pascal's principle
   d) equation of continuity

4. As the speed of a moving fluid increases, the pressure in the fluid
   a) increases
   b) remains constant
   c) decreases
   d) may increase or decrease, depending on the viscosity

5. If the cross-sectional area of a pipe decreases, what happens to the fluid velocity?
   a) Increases
   b) Decreases
   c) Remains the same
   d) Depends on the fluid density

6. A sky diver falls through the air at terminal velocity. The force of air resistance on him is
   a) half his weight
   b) equal to his weight
   c) twice his weight
   d) Cannot be determined from the information given.

7. Wind speeding up as it blows over the top of a hill
   a) Increases atmospheric pressure there.
   b) decreases atmospheric pressure there.
   c) doesn't affect atmospheric pressure there.
   d) equal's atmospheric pressure.

8. A fluid is undergoing "incompressible" flow. This means that:
   a) the pressure at a given point cannot change with time
   b) the velocity at a given point cannot change with time
   c) the velocity must be the same everywhere
   d) the pressure must be the same everywhere
   e) the density cannot change with time or location

9. A fluid is undergoing steady flow. Therefore:
   a) the velocity of any given molecule of fluid does not change
   b) the pressure does not vary from point to point
   c) the velocity at any given point does not vary with time
   d) the density does not vary from point to point

10. The equation of continuity for fluid flow can be derived from the conservation of:
   a) energy
   b) mass
   c) volume
   d) pressure

Section (B): Structured Questions

CRQ's:

1. What is difference between streamline and turbulent flow?
2. Would a drinking straw work in space where there is no gravity? Explain.
3. Why do airplanes take off into wind?
4. Describe terminal velocity in liquids.
5. Discuss the significance of Reynolds number.
6. State Bernoulli's principle.
7. Give two applications of Bernoulli's principle.
8. 'Fluid flow is turbulent rather than laminar', support this statement.
9. Discuss importance of Stokes law.
10. Justify spin of ball in Bernoulli's principle.

ERQ's:

1. Derive equation of continuity. Also show its physical significance.
2. Derive Bernoulli's equation.
3. Discuss viscous force in fluids.
4. Define fluid dynamics and explain its significance in the study of fluids. How does it differ from fluid statics?
5. Discuss the concept of Reynolds number and its significance in fluid dynamics. Explain how Reynolds number relates to the transition between laminar and turbulent flow.

Numericals:

1. Two spherical raindrops of equal size are falling through air at a velocity of 0.08 m/s. If the drops join together forming a large spherical drop, what will be the new terminal velocity? **(0.13 m/s)**

2. Calculate the viscous drag on a drop of oil of 0.1 mm radius falling through air at tis terminal velocity. (Viscosity of air = 1.8×10^{-5} Pa. s; density of oil = 850 kg / m^3) **(3.48×10^{-8} N)**

3. What area must a heating duct have if air moving 3.0 m/s along it can replenish the air every 15 minutes in a room of volume 300 m^3. Assume air density remains constant. **(0.11 m^2, 0.33 m)**

4. Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0 cm diameter pipe in the basement under a pressure of 3.0 atm, what will be the flow speed and pressure in a 2.6 cm diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches. **(1.2 m/s, 2.5 atm)**

5. What is the volume rate of flow of water from a 1.85 cm diameter faucet if the pressure head is 12 m? **(4.6×10^{-3} m^3/s)**

6. The stream of water emerging from a faucet 'neck down' as it falls. The cross-sectional area is 1.2 cm^2 and 0.35 cm^2. The two levels are separated by a vertical distance of 45 mm as shown in figure. At what rate does water flow from the tap? **(34 cm^3 / s)**

7. Water leaves the jet of a horizontal hose at 10 m/s. If the velocity of water within the hose is 0.40 m/s, calculate the pressure within the hose. Density of water is 1000 kg/m^3 and atmospheric pressure is 100000 Pa. **(1.5×10^5 Pa)**

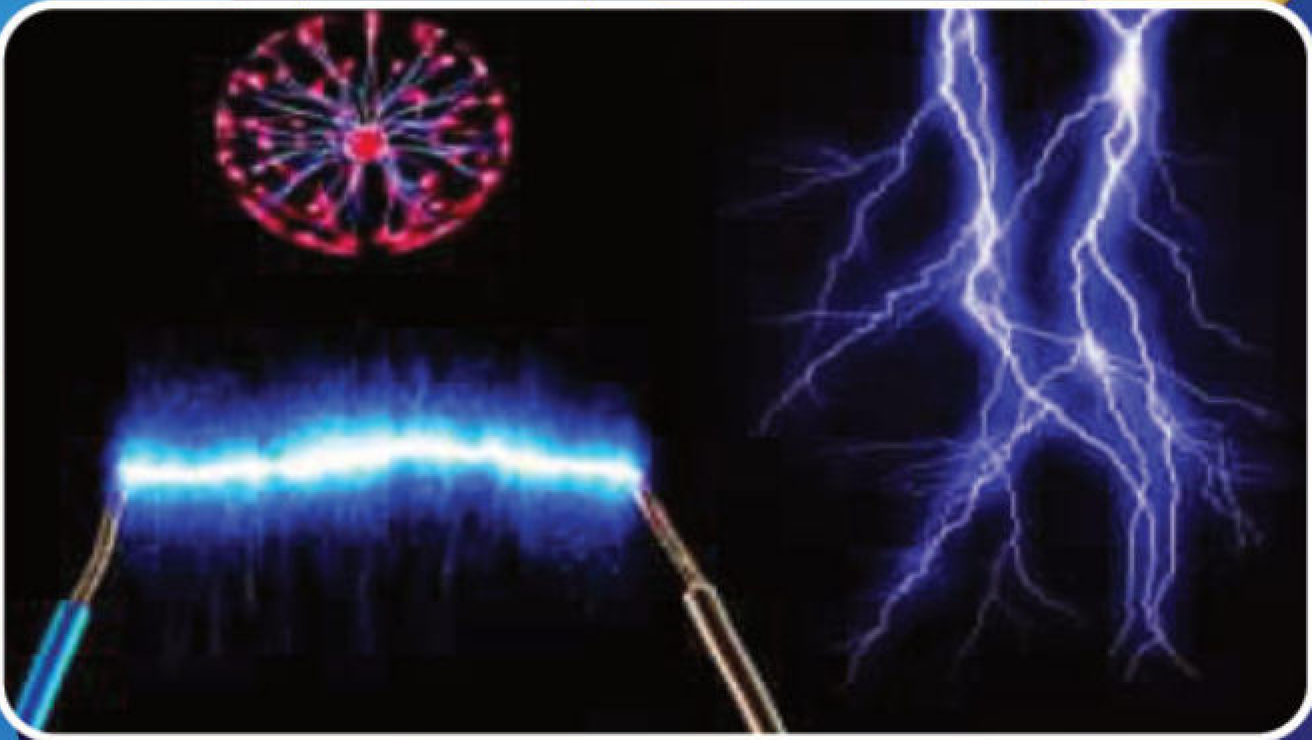
8. What is the maximum weight of an aircraft with a wing area of 50 m^2 flying horizontally, is the velocity of the air over the upper surface of the wing is 150 m/s and that the lower surface 140 m/s ? Density of air is 1.29 kg/m^3 **(9.3×10^4 N)**

9. A liquid flows through a pipe with a diameter of 0.50 m at a speed of 4.20 m/s . What is the rate of flow in L/min? **(49500 L/min)**

10. Calculate the average speed of blood flow in the major arteries of the body, which have a total cross-sectional area of about 2.1 cm^2. Use the data of example **(13.6 cm/s)**

Unit 8 Electric Fields

Teaching Periods 16 **Weightage %** 11



In this unit student should be able to:

- Define Electrostatic force
- Explain Coulomb's law
- Define the Coulomb's force in different mediums
- Solve problems using Coulomb's law
- Describe the concept of an electric field as an example of a field of force.
- Derive the expression $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$ for the magnitude of electric field at a distance 'r' from a point charge 'q'.
- Define electric field strength as a force part unit positive charge.
- Solve problems and analyze information using $\vec{E} = \frac{F}{q}$.
- Solve problems involving the use of the expression, $\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$
- Describe the concept of electric dipole
- Calculate the magnitude and direction of the electric field at a point due to two charges with the same or opposite signs.
- Sketch the electric field lines for two point-charges of equal magnitude with the same or opposite signs.
- Describe electric flux.
- Explain electric flux through a surface enclosing a charge.
- Define absolute electric potential.
- Define potential difference and its unit.
- Solve problems by using the expression V=W/q.
- Calculate the potential in the field of a point charge using the equation,
$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$$
- Show that the electric field at a point is given by the negative potential gradient at that point.
- Solve problems by using the expression, E=-V/d
- Define electron volt.

8.1.1 Electrostatic force:

In the early days, the early Greek philosophers has studied the physics of electrostatic by performing an activity; when amber is rubbed with silk or wool and brought near dust particles, the dust particles will jump to stick with the amber. This attraction between dust particles and amber is known as an electrostatic force.

The electrostatic force between two charges is analogy to the gravitational force. Two charged objects can (i) attract or (ii) repel each other with a force is known as electrostatic force.

Whereas the gravitational force is always attractive in nature. Furthermore, the electrostatic forces are short range forces as compare to the gravitational forces, but the electrostatic forces are strong forces than the gravitational forces.

The object having same nature of charges repel each other and having opposite nature attract each other with an electrostatic force.

DO YOU KNOW?

- Name: Charles-Augustin de Coulomb
- Birth Year: 1736
- Birth date: June 14, 1736
- Birth City: Angoulême
- Birth Country: France
- Gender: Male
- Best Known For: French engineer and physicist Charles de Coulomb made pioneering discoveries in electricity and magnetism, and came up with the theory called Coulomb's Law.

8.1.2 Coulomb's Law:

Charles-Augustin de Coulomb performed an experiment of torsion balance (see figure 8.1) in 1785 to measure the magnitudes of the electric force between charged objects. Once the spheres are charged by rubbing, the electric force between them is very large compared with the gravitational attraction, and so the gravitational force can be neglected. From Coulomb's experiments, we can generalize the properties of the electric force (sometimes called the electrostatic force) between two stationary charged particles.

- If two charged particles are brought near each other, they exert an electrostatic force on each other. The direction of the force vectors depends on the signs of the charges.
- If the particles have the same sign of charge, they repel each other. That means that the force vector on each is directly away from the other particle. Beside this,

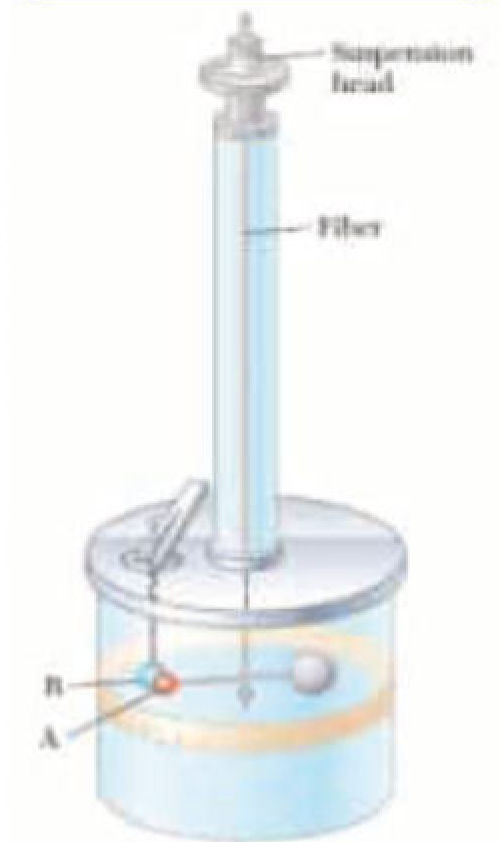


Fig: 8.1
Coulomb's torsion balance.

- If the particles have opposite signs of charge, they attract each other and the force vector on each is directly towards the other particle as shown in figure 8.2.

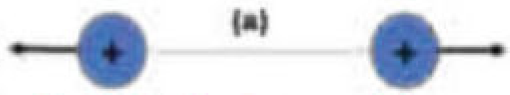


Fig: 8.2 (a) both are positive

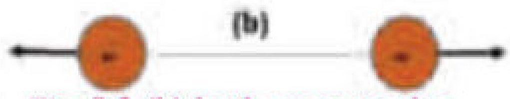


Fig: 8.2 (b) both are negative..

Fig: 8.2 (c) both are opposite

The illustration for two charges (q_1, q_2) and separation (r) between the charges is shown in figure 8.3.

Fig: 8.3
Two charges q_1 and q_2 placed at a distance r from each other.

The electrostatic force F is directly proportional to the product of magnitudes of the charges q_1 and q_2. And inversely proportional to the square of the distance r between the charges and

$$\vec{F} \propto q_1 q_2$$

$$\vec{F} \propto \frac{1}{r^2}$$

This result is known as the inverse square law.

$$\vec{F} = k\frac{q_1 q_2}{r^2}$$

Where k is known as the Coulomb's constant and $k = \frac{1}{4\pi\varepsilon_0}$.

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2} \qquad (8.1)$$

8.1.3 Coulomb's force in different medium:

The permittivity of air at standard pressure is only about 1.005 times that of ε_0. Therefore, it can be assumed in many cases that the values of ε_0 are equal for both vacuum and air. The permittivity of water is about eighty times that of a vacuum. Thus, the force between charges situated in water is eighty times less than if they were situated the same distance apart in a vacuum.

In general, the equation (8.1) can rewrite as

$$\vec{F} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}$$

where $\varepsilon = \varepsilon_r \varepsilon_0$ is the permittivity of the medium and ε_r is the relative permittivity.

DO YOU KNOW?

For your information

ε_0 is known as the permittivity of free space. if we suppose the charges are situated in a vacuum then the value of ε_0 is given as $\varepsilon_0 = 8.854 \times 10^{-12}\ C^2/N\,m^2$.
The value of Coulomb's constant in SI unit is given as
$k = 8.988 \times 10^9\ N\,m^2/C^2$

Material	Relative Permittivity
Vacuum	1.0000
Air	1.0006
PTFE, FEP (Teflon)	2.0
Polypropylene	2.20 to 2.28
Polystyrene	2.4 to 3.2
Wood (Oak)	3.3
Bakelite	3.5 to 6.0
Wood (Maple)	4.4
Glass	4.9 to 7.5
Wood (Birch)	5.2
Glass-Bonded Mica	6.3 to 9.3
Porcelain, Steatite	6.5

DO YOU KNOW?

Why do salt crystals dissolve in water?
A salt crystal is an ionic crystal. For example, in the case of sodium chloride (NaCl). The sodium ions (Na^+) and the chlorine ions (Cl^-) in a salt crystal are oppositely charged. The electrostatic forces between these ions hold the atoms together.. Note that Coulomb's law as stated above applies strictly to a vacuum. In terms of the force between two-point charges in air, the force is effectively the same as it would be in a vacuum.

Vector form of Coulomb's law:

Figure 8.4 illustrates the vector form of electrostatic force between two charges. The electric force F_{12} exerted by a charge q_1 on other charge q_2 is written as:

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2}\hat{r}_{12} \qquad (8.2)$$

where $\hat{r}_{12}$ is a unit vector directed from q_1 toward q_2 as shown Figure 8.4. Because the electric force obeys Newton's third law,as equal and opposite direction; that is, $\vec{F}_{12} = -\vec{F}_{21}$

Fig: 8.4
The force F_{21} exerted by q_2 on q_1 is equal in magnitude and opposite in direction to the force F_{12} exerted by q_1 on q_2.

Worked Example 8.1

The electron and proton of a hydrogen atom (**figure A**) are separated by approximately $5.3 \times 10^{-11}\ m$. Find the magnitudes of (a) the electric force and (b) the gravitational force between the two particles. (c) What is your conclusion about these forces.

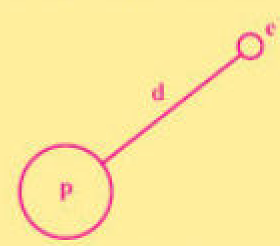


Figure A

Solution:

Step:1 Write down the known quantities and quantities to be found.

$r = 5.3 \times 10^{-11}\ m.$

(a) Charge on electron$=q_e = -1.602 \times 10^{-19} C$,
Charge on proton$=q_p = 1.602 \times 10^{-19} C$,
$k = 8.988 \times 10^9\ \ N\,m^2/C^2$
Electrostatic force$=F_e$=?

(b) Mass of electron $= m_e = 9.109 \times 10^{-31}\ kg$
Mass of proton $= m_p = 1.672 \times 10^{-27}\ kg.$
where $G = 6.67 \times 10^{-11}\ \ N\,m^2/kg^2$,
Gravitational force$=F_g$=?

Step:2 Write down the formula and rearrange if necessary.

(a) For electrostatic force $F_e = k\frac{q_e q_p}{r^2}$

(b) For gravitational force $F_g = G\frac{m_e m_p}{r^2}$

Step:3 Put the values in formula and calculate.

(a) $F_e = 8.988 \times 10^9 \ N\,m^2/C^2 \times \frac{-1.602\times 10^{-19} C \times 1.602\times 10^{-19} C}{(5.3 \times 10^{-11}\ m)^2}$

$$F_e = -8.2 \times 10^{-8} N$$

where negative sign indicates the force is attractive. The magnitude of this electrostatic force is $8.2 \times 10^{-8} N$.

(b) $F_g = 6.67 \times 10^{-11} \ N\,m^2/kg^2 \frac{9.109\times 10^{-31}\ kg \times 1.672\times 10^{-27}\ kg}{(5.3 \times 10^{-11}\ m)^2}$

$$F_g = 3.6 \times 10^{-47} N$$

Hence, the gravitational force of attraction between the particles is $3.6 \times 10^{-47} N$

(c) Conclusion:

The gravitational force between electron and proton is negligible as compared to the electric force, which implies that electric force is strong force.

Worked Example 8.2

Two positive charge of equal magnitude are placed of in vacuum at distance of 50cm and repel each other with the electric force of 0.1N. **(a)** Find the value of the charge. **(b)** Calculate the force between these two charges if they are places in an insulating liquid whose permittivity is five times that of a vacuum.

Solution:

Step:1 Write down the known quantities and quantities to be found.

$r = 0.5m$

$F = 0.1N$

$\varepsilon = 5\varepsilon_0$

(a) $q_1 = q_2 = q = ?$

(b) $F_{liquid} = ?$

Step:2 Write down the formula and rearrange if necessary.

(a) $F = k\frac{q_1 q_2}{r^2} \quad \Rightarrow \quad q^2 = \frac{F \times r^2}{k}$

(b) $F_{liquid} = \frac{1}{4\pi\varepsilon}\frac{q_1 q_2}{r^2} = \frac{1}{5\times 4\pi\varepsilon_0}\frac{q^2}{r^2}$

$F_{liquid} = \frac{1}{5}F$

Step:3 Put the values in formula and calculate.

(a) $q^2 = \frac{0.1\times(0.5)^2}{8.988\times 10^9}$

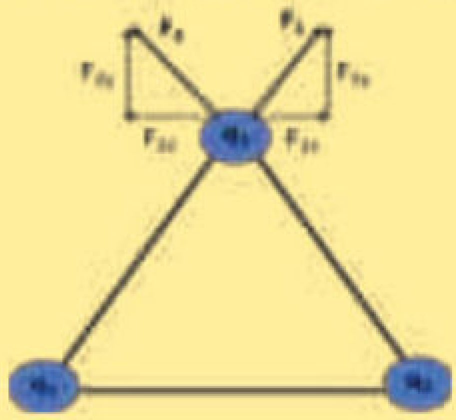
$q = 1.7 \times 10^{-7} C = 0.17\mu C$

(b) $F_{liquid} = \frac{0.1N}{5} = 0.02N$

Hence, the magnitude of electric force for an insulating liquid is $0.02N$.

Worked Example 8.3

Three charges $q_1 = +2\,\mu C$, $q_2 = +3\,\mu C$, and $q_3 = +4\,\mu C$ are placed in air at the vertices of an equilateral triangle of sides *10 cm* (see figure). Calculate the magnitude of the resultant force acting on the charge q_3?



Solution:

Step:1 Write down the known quantities and quantities to be found.

$q_1 = +2\,\mu C$, $q_2 = +3\,\mu C$, $q_3 = +4\,\mu C$, $r = 0.1\,m$ and $\theta = 60^0$

Magnitude of the resultant force acting on charge $q_3 = F_{resultant}$ =?

Step:2 Write down the formula and rearrange if necessary.

Force between the charge q_1 and q_3 $F_1 = k\frac{q_1 q_3}{r^2}$

Force between the charge q_2 and q_3 $F_2 = k\frac{q_2 q_3}{r^2}$

Now, resolving the forces F_1 and F_2 into their components and the resultant value of x and y-components (see figure 8.6)

$$F_x = F_1\cos\theta - F_2\cos\theta$$
$$F_y = F_1\sin\theta + F_2\sin\theta$$

The magnitude of the resultant force acting.

$$F = \sqrt{F_x{}^2 + F_y{}^2} = \sqrt{F_1{}^2 + F_2{}^2 + 2F_1F_2(\sin^2\theta - \cos^2\theta)}$$

Step:3 Put the values in formula and calculate.

To find the force between the charge q_1 and q_3

$$F_1 = 8.988 \times 10^9\ N\,m^2/C^2 \times \frac{2\times 10^{-6}C \times 4\times 10^{-6}C}{(0.1\,m)^2}$$
$$F_1 = 7.2\,N$$

Similarly, the force between the charge q_2 and q_3

$$F_2 = 8.988 \times 10^9\ N\,m^2/C^2 \times \frac{3\times 10^{-6}C \times 4\times 10^{-6}C}{(0.1\,m)^2}$$
$$F_2 = 10.8\,N$$

The resultant force.

$$F = \sqrt{(7.2)^2 + (10.8)^2 + 2(7.2)(10.8)\{(\sin 60^0)^2 - (\cos 60^0)^2\}}$$
$$F = 15.69\,N$$

Hence, the magnitude of the resultant force acting on the charge q_3 is 15.69N.

Self-Assessment Questions:

1. Calculate the separation between two electrons (in vacuum) for which the electric force between them is equal to the gravitation force on one of them at the earth surface.
2. If a dielectric medium of dielectric constant ε_r is filled in between the charges which are placed at distance r, then show that force between the charges decreases as compare the force when these charge place in air at same distance.

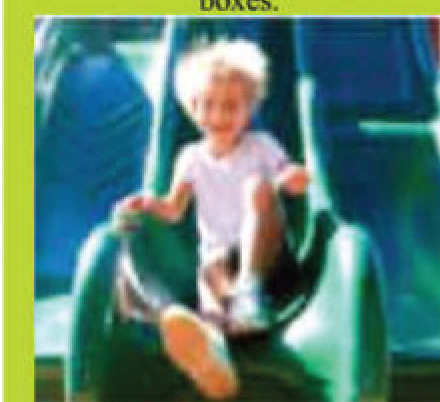
8.2.1 Electric field:

The concept of a field was developed by Michael Faraday (1791–1867) in the context of electric forces.

an electric field is said to exist in the region of space around a charged object. The charged object which produce the electric field is known as the source charge. Michael Faraday also introduced the idea of the electric field lines. These electric lines are imaginary, but these lines can be visualized by the motion of the test charge. The electric field lines around charged objects are very helpful in understanding the field strength and direction of electric field. These field lines are radial and cannot intersect each other. These field lines are originated from a positive charge and terminated on a negative charge as shown in figure 8.5. The electric field is strong where these lines are close together.

DO YOU KNOW?

To eliminate stray electric field interference, circuits of sensitive electronic devices such as T.V and computers are often enclosed with metal boxes.



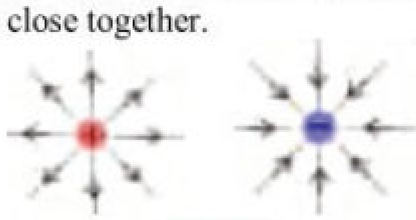


Fig: 8.5

The electric lines of force for positive and negative charge.

Strength of Electric field:

To observe the strength of electric field of the source charge, another charged object is required which is known as the test charge. This test charge experiences the electric force due to the field produced by the source charge.

To be more specific, the electric field $\vec{E}$ at a point in space around the source charge is defined as the electric force F acting on a positive test charge placed at that point divided by the test charge q_0.

$$\vec{E} = \frac{\vec{F}}{q_0} \qquad (8.3)$$

The SI unit of electric field is volts per meter (V/m) or, equivalently, newtons per coulomb (N/C).